Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/20/2006
PI Name:	Nagourney, Warren Ph.D.		
Project Title:	Ultrahigh Resolution Optical Frequency Standard Using Trapped Atoms and Single Ions		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98195	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	04/01/2004	**End Date:**	01/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dehmelt, Hans (University of Washington) Fortson, Edward (University of Washington)		
Grant/Contract No.:	GSN4495		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	We propose to construct an optical frequency standard based upon a single, laser-cooled, trapped indium ion. The proposed standard, whose absolute accuracy should approach one part in 10^18, would benefit from advances in laser stabilization technology and from the increasing compactness and efficiency of narrowband sources of radiation in the UV region of the spectrum. Recent advances in "flywheel-laser" stabilization over long integration times and the use of femtosecond lasers to coherently relate an optical to a microwave frequency suggest that the era of the optical frequency/time standard has finally arrived. The advantages of optical standards with their much higher intrinsic Q and potentially smaller systematic errors (when compared to microwave standards) can be realized using a single, laser-cooled indium ion. The task for the first year of the project will be to improve the linewidth of the recently-observed resonances of the forbidden transition to several 10's of Hz and the second year's task will be the construction of a compact, solid-state frequency-quadrupled cooling laser, a "twin" of the existing compact "clock" laser currently being used for spectroscopy. This grant is devoted mainly to the development of solid-state laser sources for the experiment: with two solid-state laser systems, the entire optical frequency standard could be quite compact and would consume a fairly small amount of electrical power; these qualities would be very favorable for the potential use of the standard in space. An extremely stable optical frequency standard has many possible applications in the areas of metrology, tests of general relativity and the equivalence principle, detection of gravitational waves as well as the more mundane areas of highly precise navigation and surveying. A frequency standard based upon a single indium ion could be used in a proposed experiment to measure the time rate of change of the fine structure constant. This has become particularly interesting due to the recent astronomical observation of a statistically significant change in alpha over a large fraction of the size of the universe. The proposed experiment would compare a single ion indium standard (the reference) to a single ion barium standard, whose change of frequency with alpha is about 10 times greater than in indium. The sensitivity in the fractional change of alpha is anticipated to be 5x10^-19/year.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved time metrology will have many benefits for navigation, surveying and communication.
Task Progress:	We replaced the somewhat unstable high-power laser-diode-based MOPA with an extremely frequency stable monolithic YAG non-planar ring oscillator (NPRO) whose output is at 946~nm. This laser is frequency quadrupled and used to drive the extremely narrow "clock" transition in single indium ions. Even though YAG oscillates at a limited number of discrete wavelengths, by a fortuitous circumstance, one of its lines (946~nm) is the exact fourth subharmonic of the "clock" transition in indium. This laser has sufficient power (about 150-300 mW) to generate a useful amount of power at 236 nm and it has an extraordinary 2~kHz free-running linewidth: more than 1000 times less than that of the MOPA! Moreover, it has convenient and fast (time constant 1~sec) temperature tuning for large frequency changes and a fast (resonant frequency 100~kHz) PZT for small frequency changes (and laser stabilization). The laser was purchased and introduced into our system early in 2004. Using a very simple feedback system and our highly-stable Zerodur optical cavity (finesse 30,000, drift 5~Hz/sec), we were able to immediately narrow the laser to under 100~Hz in the short term with a long-term drift of 5~Hz per second. This linewidth should be reduced considerably in the near future by modest improvements to the servo electronics and replacement of the Zerodur cavity by a somewhat more stable and higher-finesse cavity based upon ULE glass (recently ordered). The completely solid-state cooling laser has been completed and works extremely well. We now trap single indium ions and both observe/cool them and perform spectroscopy using solid-state lasers (all of the ion lasers and dye lasers have recently been ``banished' from our laboratory!) The cooling laser consists of a low power and quite stable oscillator laser at 922~nm which drives a tapered amplifier to produce about 400~mW of power. This radiation is then frequency quadrupled in two stages using build-up cavity sytems which are essentially identical to those used for the clock laser. A sample of the blue radiation at 461~nm is sent to a saturated tellurium spectrometer and the laser is locked to one of the tellurium resonances. The tellurium spectrometer employs an acousto-optical modulator which allows us to tune the locked laser over the ion cooling resonance.
Bibliography Type:	Description: (Last Updated: 03/30/2017)